Shashwat Joglekar

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Languages: English, French, Hindi

EDUCATION

University of Illinois Urbana Champaign: Rising Junior

Bachelor of Science: Physics (Grainger College of Engineering) Aug 2024-Present

GPA: 3.92

Relevant Coursework: Modern Experimental Physics, Classical Mechanics II, Electromagnetism II Quantum Physics I, Elementary Real Analysis, Intro to Computational Physics, Special Relativity, Differential Equations, Partial Differential Equations, Abstract Linear Algebra, Applied Complex Variables, Engineering Materials

RESEARCH

UIUC ZDC-RPD Project (ATLAS Collaboration with CERN)

Undergraduate Research Assistant: Professor Matthias Grosse Perdekamp Sep 2025-Present

- The study involves taking transmission measurements of unirradiated and CERN LHC irradiated silica fibers. The fused silica fibers from Heraeus are made of different materials
- By taking the transmission measurements and comparing them to radiation dose determined via FLUKA simulations, the radiation hardness of different fused silica materials can be determined. The study will help inform the chose of fused silica material in 2030 Run ATLAS HL-ZDC detectors
- Worked on methods to reduce random and systematic errors in optical transmission measurements conducted on 710-micron unirradiated and irradiated fused silica fibers
- Designed multiple fiber holders designs on Autodesk Fusion360 to be 3D-printed or machined
- Tweaked dimensions, tolerances and designs of fiber holders to fit them to experimental constraints
- Experimented with 3d printer tolerances to design tight but separable multi-part holders
- Incorporated suggestions from professor and labmates on methods and designs to reduce errors
- Tested for errors in different fiber holder designs by conducting transmission spectroscopy tests on sample fibers
- Tested for random errors utilizing repeated single orientation transmission measurements
- Tested for systematic errors due to rotation of irregular fiber faces through transmission measurements for four orientations separated by 90 degrees
- Continually identified sources of random and systematic errors for different fiber holder designs
- Created a fiber holder that reduces random errors and rotational systematic errors in half
- Developed a method utilizing refractive-index matching liquids to try to eliminate rotational systematic errors
- Developing a vertical orientation fiber holder to eliminate systematic errors and minimize random errors
- Analyzed transmission measurement data for random and systematic errors using percentage standard deviations
- Extracted relevant data to make concise presentations summarizing project developments, improvements and challenges

PROJECTS

Modelling Electron-Positron Annihilation

Computational Physics (PHYS246) Honors Project Fall 2025

- The project was a simulation of the collision of an electron and positron beam inside a particle accelerator
- Wrote code to model real electron and positron beams with gaussian variance to account for distributions in momentum and energy of particles
- Identified relevant particle parameters and utilized them along with relevant theory
- Incorporated advanced concepts like relativistic boosting and R-ratios in the simulation
- Solved and documented relativistic calculations for the interaction
- Coded functions to model electron positron annihilation and subsequent hadronization with a cos^2 distribution
- Utilized appropriate jet behavior for singular interactions to gauge the accuracy of hadronization functions and relativistic calculations in the simulation
- Plotted angular distribution and distribution of ejected particles on a circular detector
- Observed expected opposing jets with Gaussian distributions corresponding to the initial distributions

Simulating SBS (Stimulated Brillouin Scattering) in Optical Fibers

Computational Physics (PHYS246) Final Project — Fall 2025

- The project aimed to replicate the results of a Nieves et Al. paper on numerical simulations of SBS
- Another goal was understanding effects of gain coefficients on coupling and noise, along with producing numerically stable and physically reasonable results
- Bounced ideas with group partner to understand the paper
- Collaborated with group partner to determine how to go forward with developing code
- Developed code outlines my group partner could utilize
- Determined appropriate physical and simulation parameters by analyzing information given in the paper
- Determined and translated relevant equations in the paper to code for boundary conditions and envelope fields
- Utilized random gaussian walks to model thermal and laser phase noise
- Wrote a FDTD simulation utilizing the numerical algorithm provided in the paper
- Replicated study results to find expected cross coupling between noisy beams
- Observed gain in stokes field, and modelled the SBS interaction over time

Astrotech Project – Tau Beta Pi

Project Lead — Jul 2025-Present

- The Astrotech project aims to design interactive solar system (and eventually GNCS) planetarium experiences
- Led the team throughout the development of the project
- Guided the team with by determining deliverables, setting deadlines and creating meetings
- Delegated responsibilities for effective workflow
- Collaborated with the team and assisted them with finding information when necessary
- Created a flexible environment where team members influenced decisions
- Extracted GAIA Data for solar system objects (including Asteroids) and objects in the GNCS Database
- Developed a method to generate certain astronomical bodies in Blender utilizing their parameters
- Created object skins for solar system planets, asteroids and stars
- Developed a method to generate orbits utilizing parameters in Blender
- Designed a solar system simulation on blender with the sun, planets and 10k asteroids
- Connected Xbox flight simulator controllers to the simulation to add an interactive spaceship element
- Came in top 15% of visitor's favorite exhibits in UIUC Engineering Open House (Biggest university STEM fair)
- Got an average score of 9.71/10 from reviewers who responded to the feedback Google Form
- Currently developing similar simulations for the GAIA GNCS database and customized controllers
- Currently also working on upgrading current simulations by improving UI/UIX and adding satellites

Skills

Technical Skills: Research Skills, Design of Experiment (DOE), Data Analysis, Experiment Evaluation, Mathematical Modelling, 3D Printing, Experimental Component Design, Technical Writing, Video/Audio Editing, Computational Physics
Soft Skills: Critical Thinking and Problem-Solving Skills, Scientific and Creative Writing, Communication Skills, Presentation Skills, Collaborative Skills, Creative Skills, Digital Literacy, Organization and Time Management Skills, Document Editing Skills, Leadership Skills, Scientific Comprehension Skills, Adaptability
Programming Languages: Python, Mathematica, Kotlin
Software Proficiencies: Overleaf, VSCode, Google Collab, Jupyter Notebooks, Fusion360, Bambu